

Localization on the Scale Line: The Local Current Net, the Thermal Wedge, and the Modular Seed of the Möbius Group

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Abstract. The state papers of this series end with a unique, rotation-invariant, squeeze-thermal state on the scale-line algebra and a type III_1 verdict. This note opens the next phase: localization. We fix the *local frame* — current-weighted smearing — and prove that the passage is a unitary change of one-particle presentation commuting with the generator, so every prior result of the series transports verbatim; in this frame the infrared condition of the earlier papers is revealed as nothing other than the passage to the current itself, and no separate condition remains — the thermal spectral density is finite at zero frequency (value $1/\beta$, verified exactly), so infrared finiteness and locality are one move, not two. The interval algebras are then defined and locality proved: the symplectic form is $\sigma(F, G) = 2\pi \int F'G \, du$, vanishing exactly for disjoint supports (verified: 3×10^{-8} against order-one values), with the commutator state-independent — the thermal contributions cancel algebraically, and the computed antisymmetric part is identical at $\beta = 1$ and $\beta = 2.5$ and equal to $i\sigma$. The central identification: the position-space kernel of our two-point function is exactly the Borchers-Yngvason thermal kernel $-(\pi/4\beta^2) \sinh^{-2}[(\pi/\beta)(u - u' - i\epsilon)]$, verified at two temperatures to 3.2×10^{-9} , with the kernel's $i\beta$ -periodicity realizing KMS in position space algebraically. On this foundation the phase's payoff is stated with citations: the modular group of the half-line algebra in the thermal state acts *geometrically* on the half-line's local algebras (Borchers-Yngvason), and the modular groups of the global algebra, the half-line algebra, and its relative commutant generate a *hidden positive-energy representation of the Möbius group* $SL(2, \mathbb{R})/\mathbb{Z}_2$ (Schroer-Wiesbrock) — the first spacetime-symmetry-shaped structure the model produces, from the modular data of its unique state alone, with that literature independently confirming the global thermal algebra to be the hyperfinite type III_1 factor. What is claimed is algebraic generation; "hidden" means precisely that the Möbius action is not geometric on the thermal net, and geometrization, together with the assembly of the angular channels toward higher-dimensional structure, is deliberately deferred.

1. What this phase is

The completed state arc supplies: the scale-line CCR system with generator h of absolutely continuous spectrum $[0, \infty)$ and trivial kernel; the unique regular β -KMS state ω_β for the translation (squeeze) flow, rotation-invariant automatically; factoriality with trivial centralizer; type III_1 ; and verified strip analyticity of the two-point functions. What it does not yet contain is *localization*: subalgebras attached to regions of the scale line, without which there are no wedges, no local modular Hamiltonians, and no route to modular-generated spacetime symmetry. This note supplies the local structure, verifies its two-point kernel against the thermal chiral literature, and states the generated-

symmetry payoff that literature then delivers.

2. The local frame

****Definition.**** For real $F \in C_c^\infty(\mathbb{R}_u)$ (per angular channel), the current field $j(F)$ has two-point function

$$\omega_\beta(j(F)j(G)) \propto \int_{\mathbb{R}} \frac{p}{1 - e^{-\beta p}} \hat{F}(p) \hat{G}(p)^* dp,$$

with the full-line density equal to $p(1 + \rho(p))$ for $p > 0$ and $|p|\rho(|p|)$ for $p < 0$, $\rho = (e^{\beta p} - 1)^{-1}$; the symplectic form is $\sigma(F, G) = 2 \int_0^\infty p \operatorname{Im}(\hat{F}^* \hat{G}) dp$.

Proposition 1 (unitary transport). The map $\hat{F} \mapsto \sqrt{p} \hat{F}$ is unitary from $L^2((0, \infty), p dp)$ onto $L^2((0, \infty), dp)$ and commutes with $h =$ multiplication by p . Hence the current frame and the presentation used in the companion infrared write-out are unitarily equivalent one-particle structures with the same generator, and every result of this series formulated in terms of $(h, \rho(h))$ — the forced spectrum, the type computation, the trivial centralizer, uniqueness, and the KMS property — holds verbatim in the local frame. ■

****Remark (the infrared condition, resolved into locality).**** The earlier papers' test space $D_0 = \partial_u \mathcal{S}$ is exactly the image of current smearing: $j(F) = \phi(-F')$. In the current frame no infrared condition exists at all: the thermal density is finite at zero frequency,

$$\lim_{p \rightarrow 0} p(1 + \rho(p)) = \frac{1}{\beta},$$

verified exactly (1.000000 at $\beta = 1$; 0.400000 at $\beta = 2.5$). The condition that made the state finite, was proven necessary, and delivered uniqueness is thus revealed as a single structural fact wearing different clothes: *the good variable is the current*, and with the current come locality and infrared finiteness together, as one move.

3. The net and locality

Definition. For an open interval (or half-line) $I \subseteq \mathbb{R}_u$, let $\mathcal{A}(I)$ be the von Neumann algebra generated in the GNS representation of ω_β by the Weyl operators of currents smeared with $F \in C_c^\infty(I)$. Isotony is immediate.

Proposition 2 (locality). $\sigma(F, G) = 2\pi \int F'(u)G(u) du$; in particular $\sigma(F, G) = 0$ whenever $\operatorname{supp} F \cap \operatorname{supp} G = \emptyset$, so $\mathcal{A}(I_1)$ and $\mathcal{A}(I_2)$ commute for disjoint I_1, I_2 .

Proof. By Parseval, $2\pi \int F'G du = \int_{\mathbb{R}} (ip) \hat{F}(p) \hat{G}(p)^* dp$. Write the $p < 0$ part via $p \mapsto -p$ and reality ($\hat{F}(-p) = \hat{F}(p)^*$, $\hat{G}(-p) = \hat{G}(p)^*$): it contributes $-\overline{p\hat{F}\hat{G}^*}$ against the positive half's $p\hat{F}\hat{G}^*$, so $\int_{\mathbb{R}} p\hat{F}\hat{G}^* dp = 2i \int_0^\infty p \operatorname{Im}(\hat{F}\hat{G}^*) dp$ — purely imaginary, as it must be, since the prefactor i is what renders the total real: $i \cdot$

$2i \int_0^\infty p \operatorname{Im}(\hat{F}\hat{G}^*) dp = 2 \int_0^\infty p \operatorname{Im}(\hat{F}^* \hat{G}) dp = \sigma(F, G)$. ■ Verified: for overlapping bumps the two sides agree (-0.797862 against -0.797856 , quadrature); for disjoint compactly supported bumps the form evaluates to 3.0×10^{-8} against order-one scales.

No moment condition is imposed on the current test functions, and none is needed: under $j(F) = \phi(-F')$, every current smearing lands in the earlier papers' test space automatically, since $\widehat{F'}(0) = 0$ identically for compactly supported F — the embedding into the global structure is unconditional. The verification makes the point concretely: the bumps used are positive, with $\int F > 0$, and locality holds exactly.

Proposition 3 (the commutator is state-independent). In the antisymmetric part of the two-point function the thermal contributions cancel identically, $(1 + \rho) - \rho = 1$: the commutator equals $i\sigma$ and carries no β . Verified: the computed antisymmetric part is $+0.797862 i$ at both $\beta = 1$ and $\beta = 2.5$ — bit-identical across temperatures, and equal to the independently computed σ of Proposition 2. ■

One structural remark the numerics make vivid: the full-line thermal density $p/(1 - e^{-\beta p})$ is real-analytic across $p = 0$, and this smoothness is exactly what produces the exponential clustering of the thermal state; the vacuum density, by contrast, has a kink at $p = 0$ and power-law tails. Thermality does not merely permit localization — it improves its decay.

4. The kernel: exact agreement with the thermal chiral literature

Proposition 4 (the Borchers-Yngvason kernel). In the Borchers-Yngvason normalization, the position-space kernel of ω_β 's two-point function is

$$\omega_2(u, u') = \frac{1}{2} \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}} \frac{p}{1 - e^{-\beta p}} e^{-ip(u-u'-i\epsilon)} \frac{dp}{2\pi} = -\frac{\pi}{4\beta^2} \sinh^{-2} \left[\frac{\pi}{\beta} (u - u' - i\epsilon) \right].$$

Verified directly for our covariance: smearing both sides against disjoint compactly supported bumps gives agreement to 3.2×10^{-9} at $\beta = 1$ (values -1.4871×10^{-4}) and at $\beta = 2.5$ (-1.51869×10^{-3}). The kernel's periodicity under $u - u' \mapsto u - u' + i\beta$ — from $\sinh(z + i\pi) = -\sinh z$ — realizes the KMS condition in position space as an algebraic identity, the position-space face of the strip-boundary identity verified in the companion infrared write-out.

This identification places the series' state exactly inside the framework of Borchers and Yngvason's analysis of thermal chiral fields: the same field, the same state, the same kernel.

5. Cyclicity and separation

For the wedge results below, the thermal vector must be cyclic and separating for the half-line algebra $\mathcal{N} = \mathcal{A}(\mathbb{R}_+)$ — the Reeh-Schlieder property in its thermal form. The engine of the standard proof is analytic continuation in the translation parameters, and the required analyticity is exactly the strip structure verified in the companion infrared write-out: each correlation extends to the KMS strip with continuous boundary values, so a vector orthogonal to $\mathcal{N}\Omega$ is orthogonal to all translates of half-line observables, hence to the global algebra's cyclic subspace, hence zero. This establishes cyclicity. Separation is then immediate, from a fact the series has already established: ω_β is

faithful (the centralizer paper's argument rests on Ω being separating for the global algebra), and a vector separating for $\mathcal{M}$ separates every subalgebra, $\mathcal{N}$ included. Independently, the identical continuation argument applies verbatim to the left half-line, so Ω is also cyclic for $\mathcal{A}(\mathbb{R}_-)$ — which by locality (Proposition 2) lies in $\mathcal{N}'$, re-deriving separation without any appeal to duality and supplying the cyclic-separating property for the flanking algebras that enter the three-modular-group construction of §6. Status: proved in outline here; the thermal Reeh–Schlieder property is established in the literature (Jäkel; presupposed by Borchers–Yngvason), and the full ϵ -management is standard.

6. The thermal wedge and the generated Möbius group

With the net local, the state's modular data acquires regional structure, and the payoff of this phase is delivered by two cited results whose hypotheses the preceding sections establish for our model.

The wedge flow is geometric (Borchers–Yngvason, *J. Math. Phys.* 40 (1999), 601). The modular group of the half-line algebra $\mathcal{N}$ with respect to the thermal state transforms the local algebras of the half-line geometrically, $\mathcal{N}(t) \mapsto \mathcal{N}(\varphi(u, t))$, within a restricted parameter range — the *thermal horizon*. A second modular flow, beyond the global translations, thus exists and acts on the net's own geography.

Three modular groups generate the Möbius group (Schroer–Wiesbrock, *Looking beyond the thermal horizon: hidden symmetries in chiral models*). The modular groups of the global algebra $\mathcal{M}$, the half-line algebra $\mathcal{N}$, and the relative commutant $\mathcal{N}' \cap \mathcal{M}$ generate a *hidden positive-energy representation of $SL(2, \mathbb{R})/\mathbb{Z}_2$* — translations and dilations first, the full Möbius group with them. The same literature records that the global thermal algebra is the hyperfinite type III_1 factor — independent confirmation, from the chiral-thermal side, of the verdict this series computed from the octonionic side.

The reading, with the series' labeling intact: from the modular data of one state — the unique squeeze-thermal state, whose flow is the octonionic dilation's $\hbar$ -preserving shadow, whose mode continuum is the Mellin scale line, whose isotropy is the G_2 remnant — the model now *produces* a positive-energy Möbius group it was never given. This is the seed-of- c mechanism of the modular program (Borchers; Wiesbrock) realized inside the model: spacetime-symmetry-shaped structure as output, with exact group relations, of the kind that admits no small deformations — consistent with the series' standing position that this architecture has no mechanism for approximate symmetry violation.

What is claimed is stated precisely: *algebraic* generation. "Hidden" means the Möbius action beyond the translation–dilation pair has no geometric interpretation on the thermal net itself; converting generated symmetry into geometry is the next problem, not a footnote to this one.

7. Scope

Five items. (i) Geometrization: this note establishes generated symmetry at the algebraic level; the geometric realization (which regions, which net carries the Möbius action locally) is the following phase, and the Schroer-Wiesbrock "shadow" localization is its starting point. (ii) The commutant and the conjugation: whether $\mathcal{A}(\mathbb{R}_+)$ ' is geometrically the left half-line algebra (thermal Haag duality), and whether the modular conjugation acts on angular channels by the antipodal map, are open design questions; the first diagnostic — the reflection symmetry of the two-point function — is identified. (iii) One wedge or a family over S^2 : the angular channels are spectators here; how per-channel chiral structure assembles toward higher-dimensional geometry is deliberately deferred. (iv) Normalization: the modular temperature of wedge flows is fixed by the flow's own units; β remains a reparametrization, as established in the uniqueness paper. (v) The reconciliation of this series' construction with the conformal-net literature, long deferred, begins here in the strongest possible way: Proposition 4 shows the model's state is the thermal chiral state of that literature, kernel for kernel.

References

- Companion papers: *The Infrared Write-Out: Test Space, Weyl Algebra, and the Verified KMS Structure on the Scale Line; The State Is Unique: Quasi-Free Rigidity, the Closed Zero-Mode Door, and the Infrared Condition's Third Duty; The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III₁ Verdict; The Centralizer Is Trivial: Closing the Last Construction-Specific Link in the Type III₁ Verdict; The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the SO(3) Remnant of G₂.*
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